

API Contracts

Base URL: `http://localhost:3000/api`

Headers comunes:

```
Authorization: Bearer {token}
Content-Type: application/json
```

Modelos de Datos

User

```
{ id: string; email: string; role: 'admin' | 'teacher' | 'student'; name: string }
```

Student

```
{ id: string; name: string; email: string; career: string | null;
  semester: number | null; createdAt: Date }
```

Course

```
{ id: string; name: string; code: string; teacherId: string; credits: number;
  schedule: string; students?: string[]; totalStudents?: number }
```

Evaluation

```
{ id: string; studentId: string; courseId: string;
  type: 'exam' | 'assignment' | 'project';
  score: number; maxScore: number; date: Date; description: string }
```

TeacherSummary

```
{ id: string; name: string; email: string; role: 'teacher' }
```

DashboardMetrics

```
{ totalStudents: number; totalCourses: number; totalEvaluations: number;
  averageScore: number }
```

CoursePerformance

```
{ courseId: string; courseName: string; averageScore: number; totalStudents: number
}
```

Auth — /api/auth

POST /api/auth/login

Auth: Ninguna

Request body:

```
{ "email": "user@example.com", "password": "123456" }
```

Response 200:

```
{ "message": "...", "token": "eyJhbGciOi...", "user": User }
```

Errores: 400 campos faltantes · 401 credenciales inválidas · 500 error servidor

POST /api/auth/register

Auth: Ninguna

Request body:

```
{ "email": "user@example.com", "password": "123456", "name": "Juan Pérez", "role": "student" }
```

role: "admin" | "teacher" | "student" · password mínimo 6 caracteres

Response 201:

```
{ "message": "...", "token": "...", "user": User }
```

Errores: 400 campos faltantes / rol inválido / password corto · 500 error servidor

Students — /api/students

Método	Endpoint	Body	Response
GET	/students	—	Student[]
GET	/students/:id	—	Student
POST	/students	Partial<Student>	Student
PUT	/students/:id	Partial<Student>	Student
DELETE	/students/:id	—	void

Auth requerida: Bearer token

Campos requeridos (POST): name, email, career, semester · semester debe ser número positivo

Roles: GET → admin, teacher, student · POST/PUT/DELETE → admin

Teachers — /api/users/teachers

Método	Endpoint	Body	Response
GET	/users/teachers	—	TeacherSummary[]
POST	/users/teachers	{ name, email, password }	TeacherSummary

Auth: Bearer token · Rol: admin

password mínimo 6 caracteres

Courses — /api/courses

Método	Endpoint	Body	Response
GET	/courses	—	Course[]
GET	/courses/:id	—	Course
POST	/courses	Partial<Course>	Course

Método	Endpoint	Body	Response
PUT	/courses/:id	Partial<Course>	Course
PUT	/courses/:id/students	{ students: string[] }	Course
PUT	/courses/:id/teacher	{ teacherId: string }	Course
DELETE	/courses/:id	—	void
GET	/courses/:courseId/students	—	Student[]

Auth: Bearer token

Comportamiento GET por rol: admin → todos los cursos · teacher → sus cursos · student → cursos inscritos

PUT /:id → admin o teacher (solo sus propios cursos)

PUT /students → reemplaza la lista completa de inscritos (deduplicado automáticamente)

Campos requeridos (POST): name, code, credits, schedule · teacherId opcional

Evaluations — /api/evaluations

Método	Endpoint	Body	Response
GET	/evaluations	—	Evaluation[]
GET	/evaluations/student/:studentId	—	{ evaluations, average }
POST	/evaluations	Partial<Evaluation>	Evaluation
PUT	/evaluations/:id	Partial<Evaluation>	Evaluation
DELETE	/evaluations/:id	—	void

Auth: Actualmente sin middleware (sin token requerido)

type: "exam" | "assignment" | "project" · score no puede superar maxScore

GET /student/:id responde: { evaluations: Evaluation[], average: number }

Dashboard — /api/dashboard

Método	Endpoint	Body	Response
GET	/dashboard/metrics	—	DashboardMetrics
GET	/dashboard/performance	—	CoursePerformance[]

Auth: Sin middleware (sin token requerido)

metrics incluye: totalStudents, totalCourses, totalEvaluations, averageScore

atRiskCount = estudiantes con promedio < 60

Códigos de Error Globales

Código	Significado
400	Validación fallida / campos faltantes
401	Token ausente o inválido
403	Rol sin permisos suficientes
404	Recurso no encontrado
500	Error interno del servidor